Programming for Data Analysis and AI

Coursework

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This report will analyse a dataset generated by the World Health Organisation which investigates 5110 patients, based on demographics such as age, gender, marital status, work type, residence type; health indicators such as BMI, average glucose levels and smoking status; and health conditions including hypertension, heart disease and stroke.

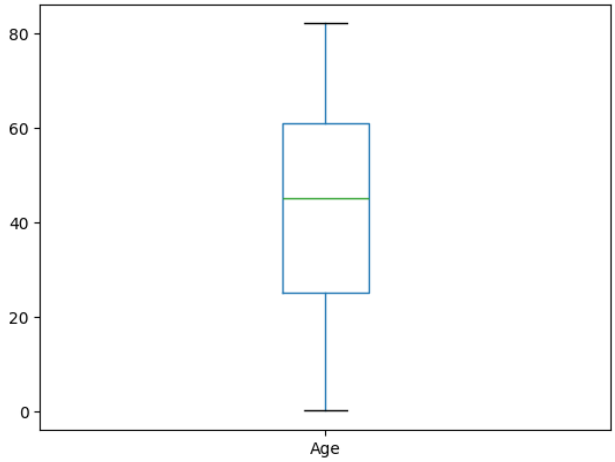
1. Demographic Information

Table 1.

Basic statistics for numerical data

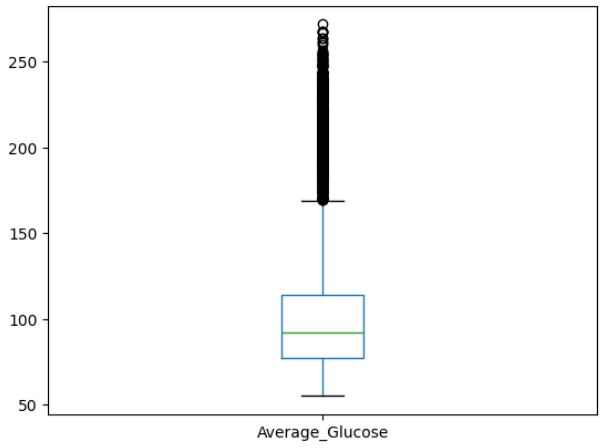
	Count	Mean	Standard Deviation	Minimum Value	Lower Quartile	Middle Quartile	Upper Quartile	Maximum Value
Age	5110	43.23	22.61	0.08	25.00	45.00	61.00	82.00
Average Glucose Level	5110	106.15	45.28	55.12	77.24	91.89	114.09	271.74
BMI	4909	28.89	7.85	10.30	23.50	28.10	33.10	97.60

Figure 1a.
Boxplot of age



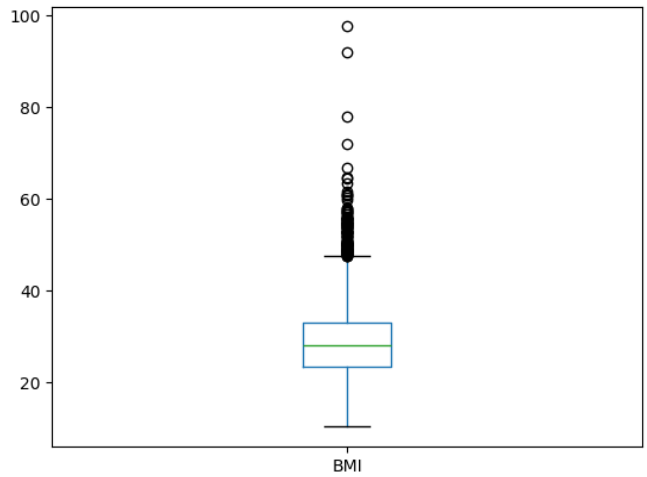
The dataset consists of 5110 age entries with the average age of respondents being approximately 43 years of age (*M = 43.23),* with a widespread range of ages (*SD = 22.61*). Ages range from 0.08 (approximately 1 month old) to 82 with a median of 45, most respondents are within 25 to 61 years of age. Age spans from infancy to old age; presenting a representative sample.

Figure 1b.
Boxplot of average glucose levels



5110 individuals recorded their average glucose levels, with an average of 106.15 mg/dL and large variation of levels across respondents (*SD = 46.28).* Reported glucose levels ranged from 55.12 to 271.74 mg/dL, with median 91.89. Majority individuals indicate normal glucose levels, while higher values suggest health concerns. As seen in Figure 1b, there are many outlier values approximately after 175mg/dl.

Figure 1c.
Boxplot of BMI



4909 BMI entries were noted, meaning there are 201 missing pieces of BMI data, with outliers appearing after an approximated BMI index of 50. The mean value is 28.89, suggesting majority of the class is overweight (healthy BMI lies between 18.5 and 24.9; WHO, 2019), with a standard deviation of 7.85, ranging from 10.30 to 97.60, indicating a lot of variances in BMI across the sample.

It was of interest to observe if there is any relationship between these variables, so a Pearson Correlation test was conducted, see Table 2.

Table 2.

Pearson correlation coefficients between numerical variables

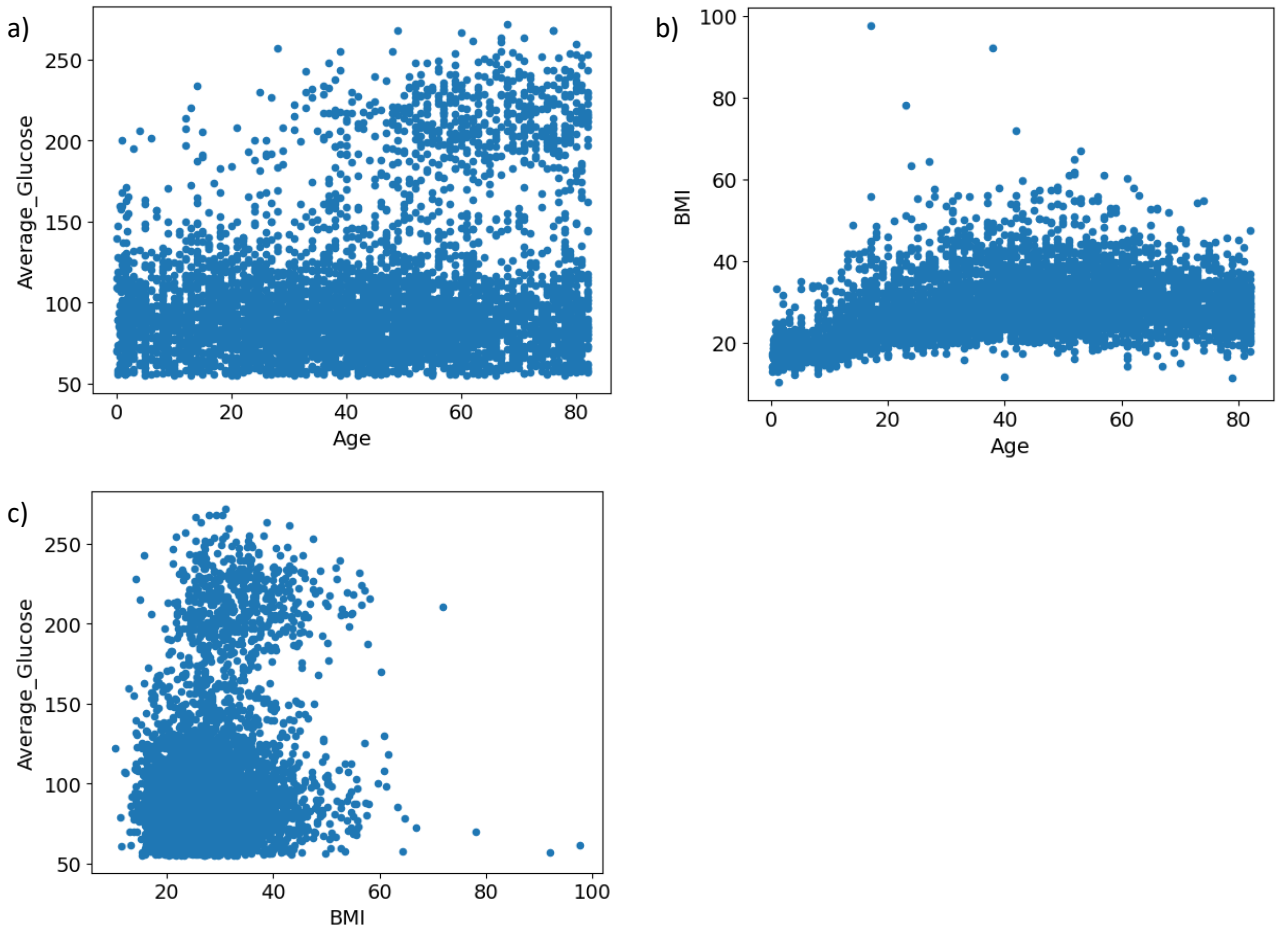
	Age	Average Glucose Levels	BMI

Age	1.00	0.24	0.33
Average Glucose Levels	0.24	1.00	0.18
BMI	0.33	0.18	1.00

For a visual representation of these values, see Figure 2.

Figure 2.

Visual representation of correlations in Table 2.



A correlation coefficient of 0.24 suggests a weak positive correlation between age and average glucose levels, Figure 2a illustrates older individuals tend to have higher average glucose levels, though this relationship is not strong, given there is no obvious linear trend. This weak trend downplays the notion that older individuals are more likely to have higher glucose levels, suggesting other factors play a role.

BMI and age express a stronger positive correlation (r = 0.33) than age and average glucose but a weak one nonetheless. Figure 2b portrays BMI increases with age, peaking at around the age of 40, then shows a slight steady decline as age increases after 40.

With r=0.18, Figure 2c further enforces a weak positive correlation between BMI and average glucose levels, implying that higher BMI values may correlate to higher average glucose levels, but the relationship is minimal.

The remaining variables in this dataset were categorical, see Table 3.

Table 3.

Basic statistics for categorical data

	Frequency		Percentage Distribution		Mode	Interpretation
Gender	Male Female Other	2115 2994 1	Male Female Other	41.39% 58.59% 0.02%	Female	Majority respondents are female
Ever married?	Yes No	3353 1757	Yes No	65.62% 34.38%	Yes	Most of the sample are or have been married
Work Type	Private Self-employed Children Government job Never Worked	2925 819 687 657 22	Private Self-employed Children Government job Never Worked	57.24% 16.03% 13.44% 12.86% 0.43%	Private	Vast majority of respondents work private jobs.
Residence Type	Urban Rural	2596 2514	Urban Rural	50.80% 49.20%	Urban	Participants are evenly distributed across urban and rural areas
Smoking Status	Never Smoked Unknown Formerly smoked Smokes	1892 1544 885 789	Never Smoked Unknown Formerly smoked Smokes	37.03% 30.22% 17.32% 15.44%	Never Smoked	Most of the sample have never smoked or their smoking status is unknown
Hypertension	No Yes	4612 498	No Yes	90.25% 9.75%	No	The vast majority do not experience hypertension
Heart Disease	No Yes	4834 276	No Yes	94.60% 5.40%	No	The sample largely does not suffer from heart disease
Stroke	No Yes	4861 249	No Yes	95.13% 4.87%	No	Most of the respondents have not experienced a stroke.

Environmental factors such as work type and marriage status can influence smoking status. A more stressful job or unsuccessful marriage can increase the likelihood of a smoker status (Kouvonen et al., 2005) , see Figure 3.

Figure 3a.

Contingency table showing the relationship between work type and smoking status

	Unknown	Formerly Smoked	Never Smoked	Smokes
Government Job	122	133	282	120
Never Worked	8	0	14	0
Private	640	531	1211	543
Self-employed	156	208	331	124
Children	618	13	54	2
Interpretation	Most individuals in private/self-employed work have never smoked, although many are former or current smokers. Government workers mainly deviate from a smoker status, and children are clustered around the 'Unknown' category, given their age possibly. The table suggests there may be a relationship between more stressful jobs and smoking, with private/self-employed workers – greater job-related stress – have higher values of former and current smoker status compared to those in government jobs.			

Figure 3b.

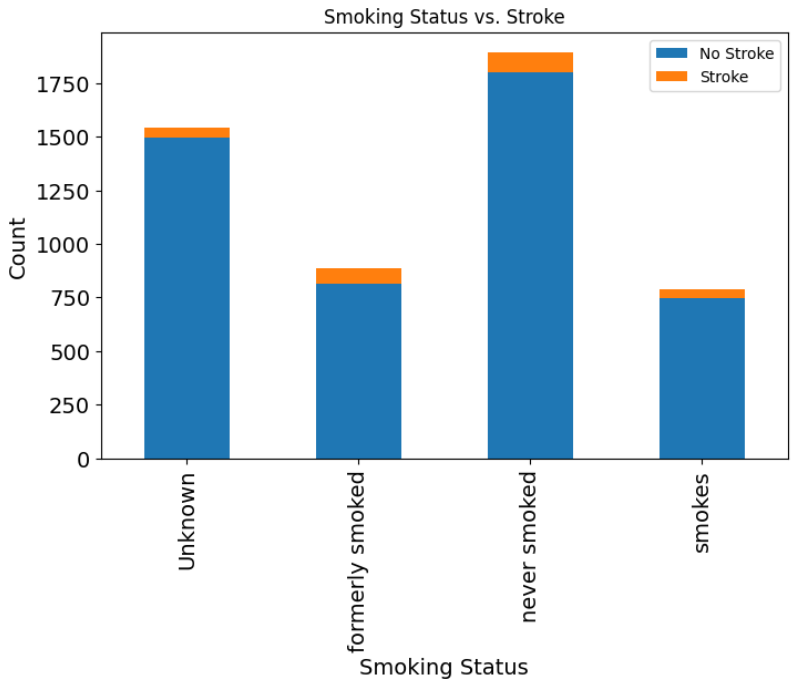
Contingency table showing the relationship between marriage status and smoking status

	Unknown	Formerly Smoked	Never Smoked	Smokes
Never Married	901	147	530	179
Have been married	643	738	1362	610
Interpretation	There is a potential relationship between these two variables. Those who have never been married likely smoke less (*N* = 179) than those that have been married (*N* = 610).			

Additionally, smoking increases the likelihood of a stroke (Shah & Cole, 2010) so it was of interest to determine if this is true in this instance, see Figure 4.

Figure 4

Bar chart representing the relationship between smoking status and stroke



There is no distinct relationship between smoking status and stroke presented in Figure 4, smokers do not seem to experience more strokes compared to those that have never smoked.

The likelihood of developing hypertension, heart disease and a stroke is differentiated between genders (Chen et al., 2019). Contingency tables and chi-squares were conducted to observe if this notion is consistent within this dataset, see Figures 5.

Figure 5a.

Contingency table for the relationship between gender and hypertension

	No	Yes
Female	2718	276
Male	1893	222
Other	1	0
Chi-Square Value	0.300	

Females appear to be more susceptible to hypertension in this dataset, although, a chi-square value of 0.300 indicates no significant relationship between gender and hypertension ($X^2 >$ 0.05).

Figure 5b.

Contingency table for the relationship between gender and heart disease

	No	Yes
Female	2881	113
Male	1952	163
Other	1	0

Chi-Square Value	$6.93e^{-09}$

Males demonstrate noticeably higher amount of heart disease compared to females, an almost miniscule chi-square value conveys a significant relationship between gender and heart disease, where males in this dataset are significantly more at risk of heart disease than females ($X^2 < 0.05$)

Figure 5c.

Contingency table for the relationship between gender and stroke

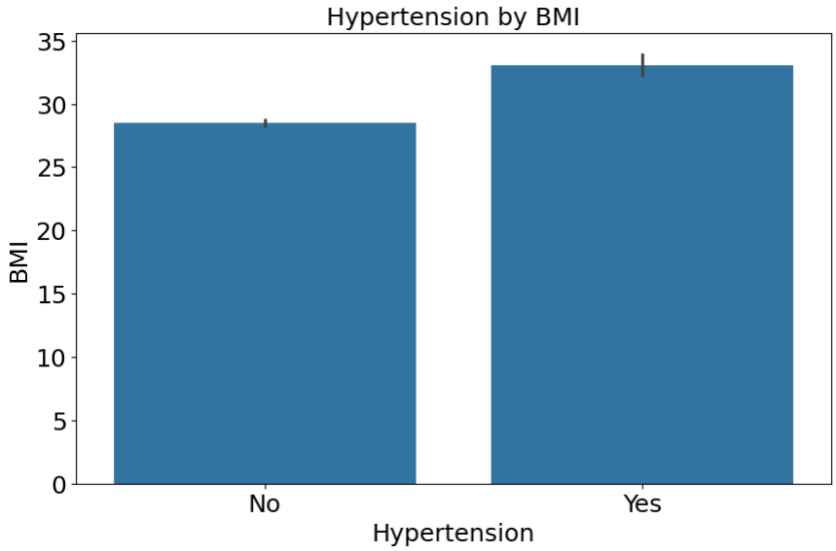
	No	Yes
Female	2853	141
Male	2007	108
Other	1	0
Chi-Square Value	0.790	

The number of females that have experienced a stroke outweighs the number of males, but the difference is not substantial. A high chi-square value reveals there is no significant difference between gender and stroke ($X^2 > 0.05$).

BMI also effects the development of stroke (Wang et al., 2022), hypertension (Jiang et al, 2016) and heart disease (Alexander, 2001). Bar charts were presented to demonstrate if there is any noticeable indication this concept is true, see Figure 6.

Figure 6a.

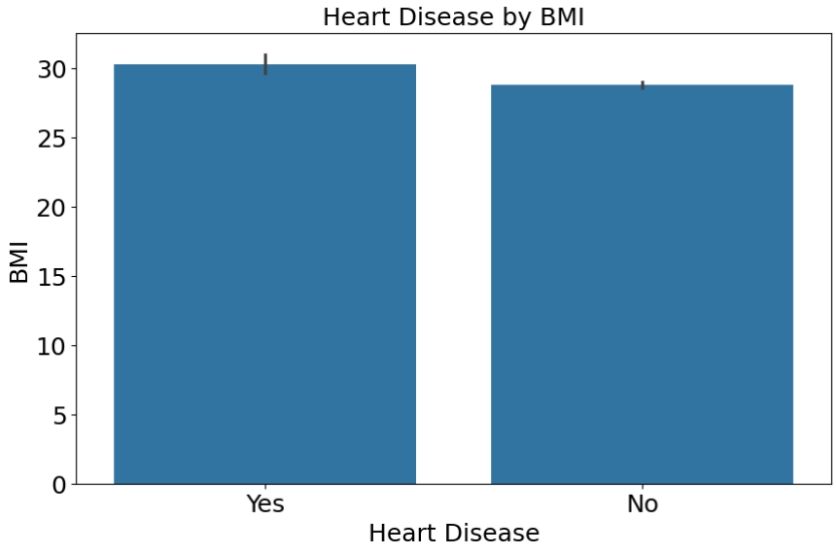
Bar chart representing the relationship between BMI and hypertension



Individuals with hypertension quite distinctly expressed a higher BMI than those without hypertension.

Figure 6b.

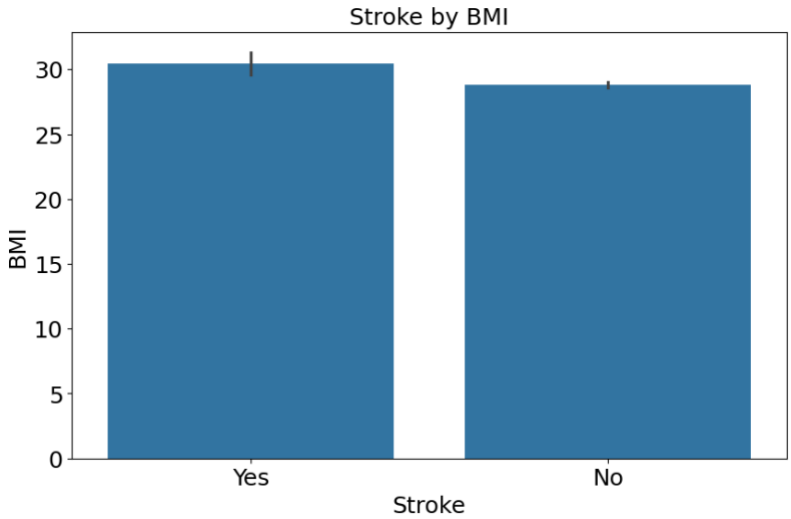
Bar chart representing the relationship between BMI and heart disease



Those with heart disease demonstrate higher BMI values than those without heart disease, but the difference is minimal.

Figure 6c.

Bar chart representing the relationship between BMI and stroke



Individuals that have experienced a stroke report higher BMI value, but the difference compared to those who have not experienced a stroke is not substantial.

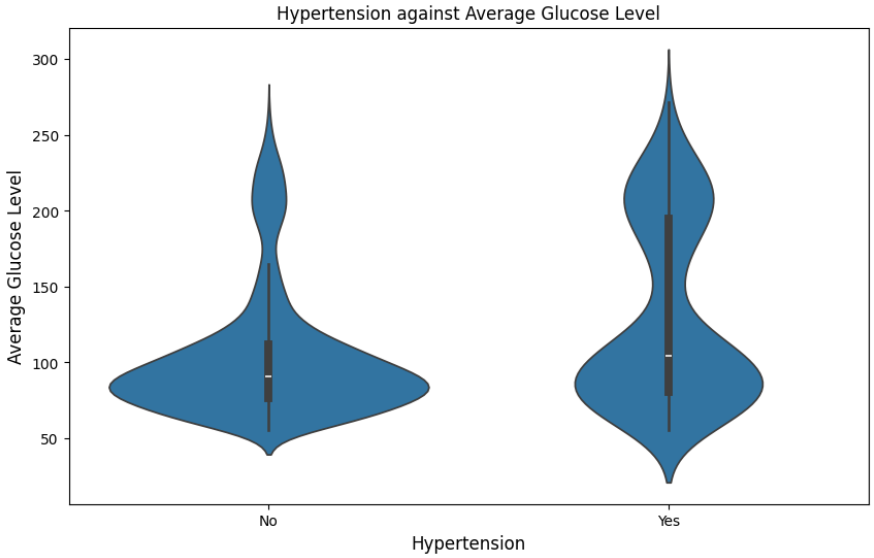
Although some differences are not extremely distinct, it is suggested individuals with hypertension, heart disease and have experienced a stroke, tend to have a higher BMI. To determine if any of these relationships are significant, a multivariate analysis of variance (MANOVA) will be used to test if there is any effect of the independent variable (BMI) on the dependent variables (hypertension, heart disease and stroke): $F(3,4905) = 48.66$, $p = 0.971$. BMI does not have a statistically significant effect on hypertension, heart disease or stroke likelihood within this dataset. Although the descriptive trends seen in Figure 6 suggest these conditions are associated with higher BMIs, the MANOVA highlights such differences are not robust

enough to determine a significant effect. Therefore, it is difficult to generalise the notion that these conditions are the product of higher BMI in real-world applications.

Higher glucose levels increase the likelihood of developing hypertension/heart disease and having a stroke (Poznyak, 2022). Violin plots and MANOVA will be used to investigate if this trend is consistent within this dataset.

Figure 7a.

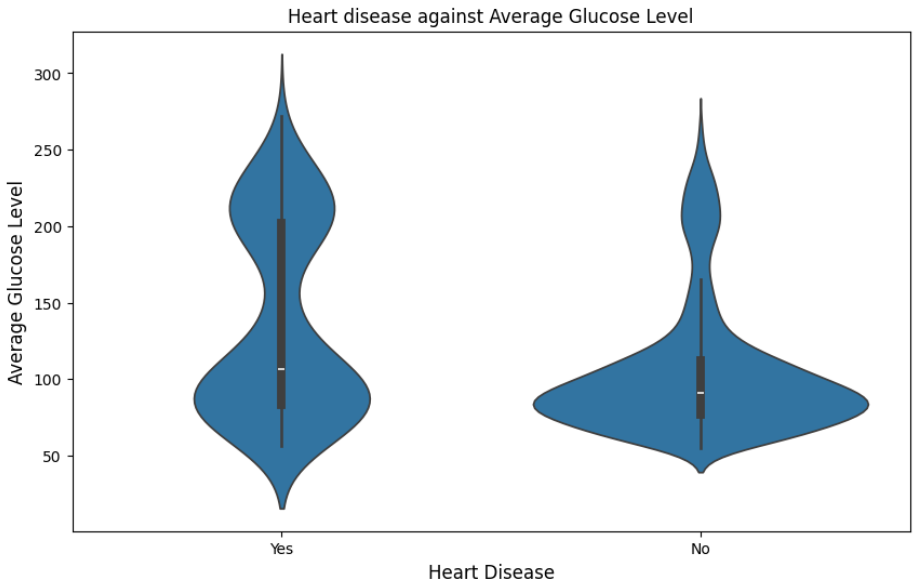
Violin plot representing the relationship between hypertension and average glucose levels



Those with hypertension demonstrate higher average glucose levels compared to those without. The distribution of hypertensive individuals is narrow with slightly larger median and interquartile range, suggesting an association between hypertension and glucose levels.

Figure 7b.

Violin plot representing the relationship between heart disease and average glucose levels



People with heart disease reported higher average glucose levels, this distribution is narrow with a slightly higher median and interquartile range, also suggesting a relationship between average glucose levels and heart disease.

Figure 7c.

Violin plot representing the relationship between stroke and average glucose level

Stroke patients reported higher glucose levels than non-stroke patients, with a higher median and interquartile range, it is possible there is an association between higher glucose levels and likelihood of experiencing a stroke.

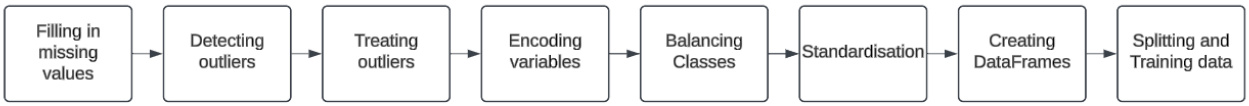
Overall, it is implied higher glucose levels can influence the likelihood of developing hypertension, heart disease and a stroke or vice versa. To determine if this relationship is significant, a multivariate analysis of variance was conducted, predicting the dependent variables (‘hypertension,’ ‘heart disease,’ ‘stroke’) will be influenced by the independent variable (‘average glucose level’): $F(3,5106) = 108.41, p = 0.940$. A high F-statistic suggests

differences in the dependent variables based on average glucose level, but this is not supported by the data given a p-value that is not statistically significant ($p > 0.05$). Once again, we cannot generalise any assumptions about these conditions being influenced by average glucose levels to the real world.

2. Data Preparation

Figure 8.

Data Preparation pipeline



2.1: Filling in missing values

Classification, regression and clustering models will primarily be used to analyse this dataset, these models only perform with complete data, hence, appropriate imputation technique is necessary. Only one variable had missing values, BMI with 201 unknown indexes. In this instance, missing values were replaced with the most common reported BMI value (28.7).

2.2 Detecting outliers

Outliers significantly deviate from the rest of the dataset, such values can skew statistical measures such as the mean and standard deviation, in turn, impacting the accuracy of standardised data in clustering models for instance. Outliers in this dataset were identified using the interquartile range (IQR) method on numerical variables (age, BMI and average glucose levels) by identifying data points that fall below: (lower quartile – 1.5) x IQR/ or above (upper quartile +1.5) x IQR, see Figure 9.

Figure 9a.

Outlier values for BMI

```
21      48.9
66      47.5
113     56.6
254     50.1
258     54.6
        ...
4906    54.1
4952    56.6
5009    49.5
5057    47.6
5103    46.9
Name: bmi, Length: 126, dtype: float64
126

number of outliers 126
max outlier value 97.6
min outlier value 10.3
```

Figure 9b.

Outlier values for Average Glucose Level

```
0       228.69
1       202.21
3       171.23
4       174.12
5       186.21
         ...
5061    223.78
5062    211.58
5063    179.38
5064    193.88
5076    174.37
Name: avg_glucose_level, Length: 627, dtype: float64
627

number of outliers 627
max outlier value 271.74
min outlier value 169.43
```

2.3 Treating outliers

As seen in Figure 1b and 1c, BMI and average glucose levels have many outlier data values. The IQR method highlighted these outliers and provide key information regarding it, see Table 4.

Table 4.

Outlier information

	Number of outliers	Minimum outlier value	Maximum outlier value
BMI	126	10.3	97.6
Average Glucose Level	627	169.43	271.74
Total	**753**		

Table 4 highlights there are 753 outliers in the dataset, to treat this, these 753 indexes were removed, reducing the sample size to 4,391 data entries from 5,110.

2.4 Encoding variables

Categorical variables were converted into numerical form using label encoding, see Table 5. Classification, regression and clustering all require numerical data; this step was essential.

Table 5

Transformed categorical data

	Before Encoding	After Encoding
Gender	Female Male	0 = Female 1 = Male
Ever Married	No Yes	0 = No 1 = Yes
Work Type	Government Job Never Worked Private Self-Employed Children	0 = Government Job 1 = Never Worked 2 = Private 3 = Self-employed 4 = Children
Residence Type	Rural Urban	0 = Rural 1 = Urban
Smoking Status	Unknown Formerly Smoked Never Smoked Smokes	0 = Unknown 1 = Formerly Smoked 2 = Never Smoked 3 = Smokes

2.5 Balancing classes

As can be seen in Table 3, there is a substantial class imbalance within the 'stroke,' 'heart disease,' and 'hypertension' variables, with the vast majority reporting they have not

experienced any of these conditions. Performing classification with imbalanced data will lead to bias in classification models performing better for majority classes than the minority. Synthetic Minority Oversampling Technique (SMOTE) will be applied to these three classes, where the minority group ('Yes') will be oversampled via generation of synthetic data, this will balance the classes.

Table 6

Class transformation

Variable	Before SMOTE		After SMOTE	
Hypertension	No	4065	No	4065
	Yes	326	Yes	4065
Heart Disease	No	4219	No	4219
	Yes	172	Yes	4219
Stroke	No	4226	No	4226
	Yes	165	Yes	4226

2.6 <u>Standardisation</u>

Standardisation was conducted after outlier treatment as outliers skew standardised statistical measures, such as mean and standard deviation. It also requires data in numerical form, hence conducted after encoding too. This step is necessary for the clustering model to ensure all attributes contribute equally, see Figure 10.

Figure 10a.

Sample of data before standardisation

gender	age	hypertension	heart_disease	ever_married
1	67.0	0	1	1
0	61.0	0	0	1
1	80.0	0	1	1
0	49.0	0	0	1
0	79.0	1	0	1

Figure 10b.

After standardisation

gender	age	hypertension	heart_disease	ever_married
1.188073	1.051434	-0.328602	4.185032	0.723884
-0.840344	0.786070	-0.328602	-0.238947	0.723884
1.188073	1.626390	-0.328602	4.185032	0.723884
-0.840344	0.255342	-0.328602	-0.238947	0.723884
-0.840344	1.582163	3.043196	-0.238947	0.723884

2.7 <u>Creating a separate dataframe to be used for regression</u>

The target attribute used for regression will be 'BMI,' so this variable needs to be moved to the end of the table.

```
df_regression = df_numerical[['id', 'gender', 'age', 'hypertension', 'heart_disease', 'ever_married',
       'work_type', 'Residence_type', 'avg_glucose_level',
       'smoking_status', 'stroke', 'bmi']]
```

2.8 Splitting and training the data for classification and regression

Classification

```
from sklearn.model_selection import train_test_split

X = df_resampled.iloc[:,1:-1].values
y = df_resampled.iloc[:,-1].values

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)
```

Regression

```
X2 = df_regression.iloc[:,1:-1].values
y2 = df_regression.iloc[:,-1].values
from sklearn.model_selection import train_test_split
X2_train, X2_test, y2_train, y2_test = train_test_split(X2, y2, test_size = 0.25, random_state = 0)

from sklearn.model_selection import train_test_split
X2_train, X2_test, y2_train, y2_test = train_test_split(X2, y2, test_size = 0.25, random_state = 0)
```

3. Classification

Classifiers

Various classification algorithms will be applied to the 'Stroke' variable in this dataset, predicting two classes: No Stroke (NS) and Stroke (S), using:

1. Decision Tree (DT)
2. Naïve Bayes (NB)
3. K-Nearest Neighbour (KNN)
4. Support Vector Machine (SVM)
5. Logistic Regression (LR)
6. AdaBoost Classification

3.1 *Decision Tree Classification*

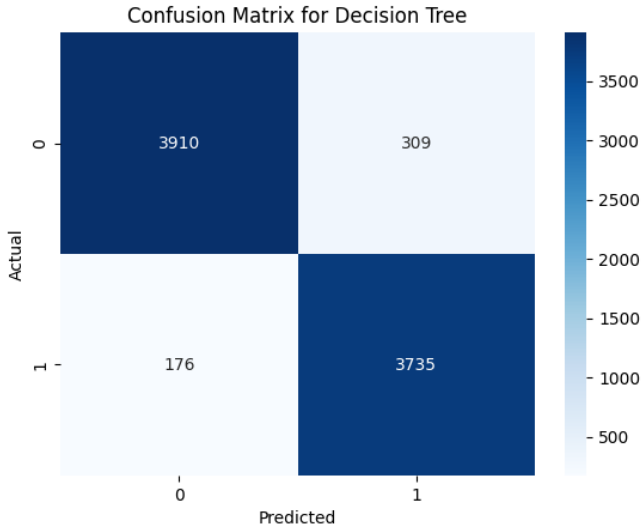
Table 7a.
Performance Metrics

	Precision	Recall	F1-score	Support	Overall Accuracy	Overall AUC
No stroke (NS)	0.96	0.93	0.94	4219	0.940	0.941
Stroke (S)	0.92	0.95	0.94	3911		
Interpretation	DT performed strongly in predicting NS and S cases, with an overall accuracy of 94.03%. There is high precision for both classes, where 96% and 92% of predictions for NS and S classes are accurate, respectively. Predicting NS slightly outperformed predictions for S classes, but the difference is very minimal. High recall for both classes suggests Decision Tree is highly sensitive to detecting both NS and S instances. An identical F1-score across both classes demonstrates a well-balanced performance					

	between precision and recall. A large AUC value of 94.1% indicates the DT model excelled in discriminatory power between both classes.

Figure 11.

Confusion Matrix visualisation and table



This model excelled in identifying true cases of NS and S, with a total of 7645 correct predictions and, although 0.06%, 485 instances, were misclassified, suggesting there is room for improvement in the Decision Tree classifier, in despite of its excellent performance.

Number of instances	True NS	False NS	False S	True S	Total Correct Predictions	% of correct predictions	Total incorrect Predictions	% of incorrect predictions
8130	3910	176	309	3735	7645	94.03%	485	5.97%

Table 7b.

Evaluation of the DT model

Strengths	**Limitations**
High Accuracy – 94.03% accuracy in predicting true classes.	**False Positives** – 309 misclassifications of 'No Stroke' cases incorrectly classified as 'Stroke.'
Balanced performance – high precision, recall, F1-score and AUC	**Potential Overfitting -** consistently high metrics suggest possible overfitting
Low false negatives – minimising misclassification of 'Stroke' cases as "No stroke' is important in health-environments, where incorrectly predicting a "stroke" variable as "No Stroke' can have severe consequences	

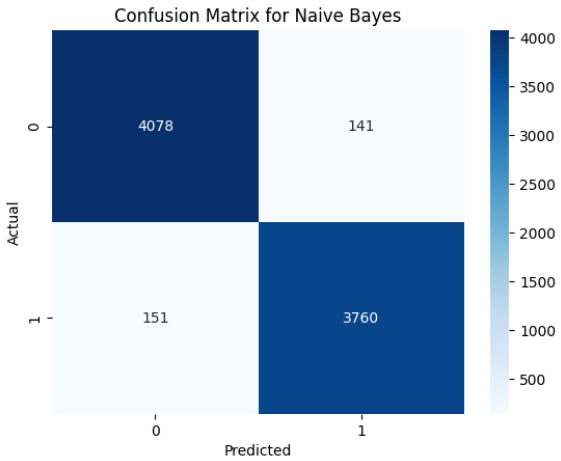
3.2 *Naïve Bayes algorithm*

Table 8a.

Performance metrics

	Precision	Recall	F1-score	Support	Overall Accuracy	Overall AUC
No stroke	0.96	0.97	0.97	4219	0.964	0.964
Stroke	0.96	0.96	0.96	3911		
Interpretation	Excellent classification performance with accuracy of 96.4% and great discrimination between both classes with an AUC of 0.964. Precision, recall and F1-scores are consistently high for S and NS classes, indicative of the model's predictive accuracy.					

Figure 12.
Confusion matrix visualisation and table



Out of 8130 cases, 96.40% of instances were correctly predicted, while 3.60% were incorrectly classified, indicating a very strong performance by the Naive Bayes algorithm.

Number of instances	True NS	False NS	False S	True S	Total Correct Predictions	% of correct predictions	Total incorrect Predictions	% of incorrect predictions
8130	4078	151	141	3760	7838	96.40%	292	3.60%

Table 8b.
Evaluation of Naïve Bayes classifier

Strengths	**Limitations**
Outperforms DT – higher accuracy (96.40%) than DT classifier (94.03%)	**False negatives** – minimising misclassification of 'Stroke' cases as "No stroke' is important in health-environments, where incorrectly predicting a "stroke" variable as "No Stroke' can have severe consequences
Balanced performance – high precision, recall, F1-score and AUC	

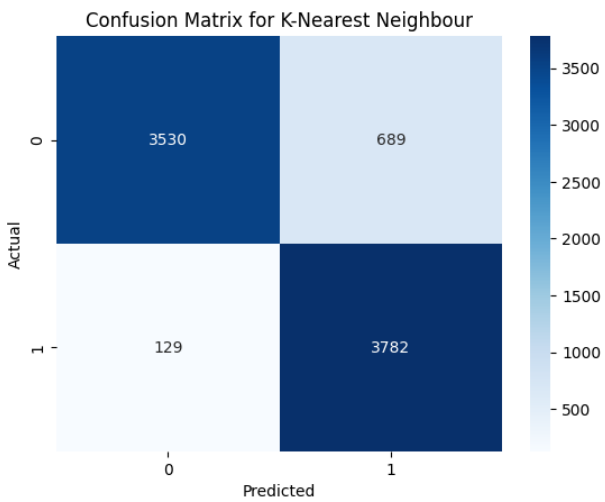
3.3 *KNN algorithm*

Table 9a.
Performance metrics

	Precision	Recall	F1-score	Support	Overall Accuracy	Overall AUC
No stroke	0.96	0.84	0.90	4219	0.90	0.90
Stroke	0.85	0.97	0.90	3911		
Interpretation	KNN classifier more accurately classified 'No Stroke' classes than 'Stroke' classes, given a higher precision score, but the model demonstrates more sensitivity to the 'Stroke' class with a higher recall score, and greater difficulty in distinguishing the 'No Stroke' class. An identical F1-score demonstrates a well-balanced performance in detecting true 'Stroke' instances while minimizing false 'Stroke' and false 'No Stroke' instances.					

Figure 13.

Confusion matrix visualisation and table



This model performed relatively the weakest, capturing 7312 correct prediction (89.94%), where 818 instances (10.06%) were misclassified, indicating this model performed the least effectively.

Number of instances	True NS	False NS	False S	True S	Total Correct Predictions	% of correct predictions	Total incorrect Predictions	% of incorrect predictions
8130	3530	129	689	3782	7312	89.94%	818	10.06%

Table 9b.

Evaluation of KNN classifier

Strengths	**Limitations**
Balanced F1-Scores – 0.90 for both classes, suggesting balanced precision and recall which reflects a fair classification performance overall.	**Lower recall for 'No Stroke' –** KNN misclassifies more NS cases compared to other classifiers
	Higher False Positives – incorrect predictions of 'No Stroke' cases as 'Stroke' will lead to unnecessary further testing and interventions for potential misdiagnosis.
	k-Value – the value selected for '*k*' affects the performance accuracy of the model,

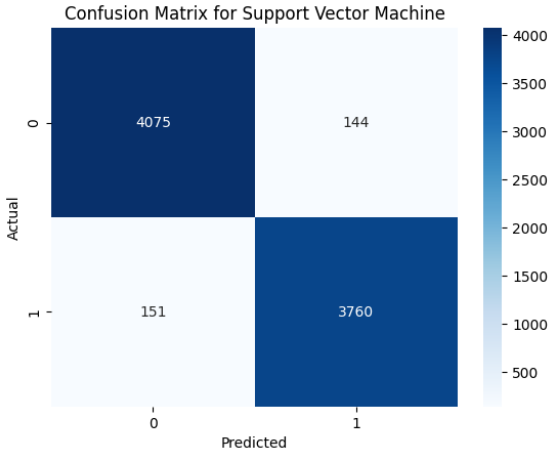
	adjusting this value may improve its performance.

3.4 *Support Vector Machine*

Table 10a.
Performance metrics

	Precision	Recall	F1-score	Support	Overall Accuracy	Overall AUC
No stroke	0.96	0.97	0.97	4219	0.96	0.96
Stroke	0.96	0.96	0.96	3911		
Interpretation	Excellent overall accuracy (96.37%) and great discriminatory power between both classes (AUC = 0.96). Precision, recall and F1-score are all consistent and high across the classes, delivering a balanced performance while minimising false 'stroke' and 'no stroke' instances.					

Figure 14.
Confusion matrix visualisation and table



The SVM model correctly predicted 7835 cases (96.37%), while 295 cases (3.63%) were misclassified. The false positives and negatives identified are low, indicating this model classified both classes efficiently and effectively.

Number of instances	True NS	False NS	False S	True S	Total Correct Predictions	% of correct predictions	Total incorrect Predictions	% of incorrect predictions
8130	4075	151	144	3760	7835	96.37%	295	3.63%

Table 10b.
Evaluation of SVM

Strengths	**Limitations**
High Precision and Recall – lower false classifications for both 'No Stroke' and 'Stroke' instances	**False negatives –** even though it is low, 151 false negatives can be critical in healthcare settings, where actual strokes are being predicted as no strokes.
Balanced Performance – minimal trade-off between recall and precision.	**Expensive –** SVM is computationally expensive.

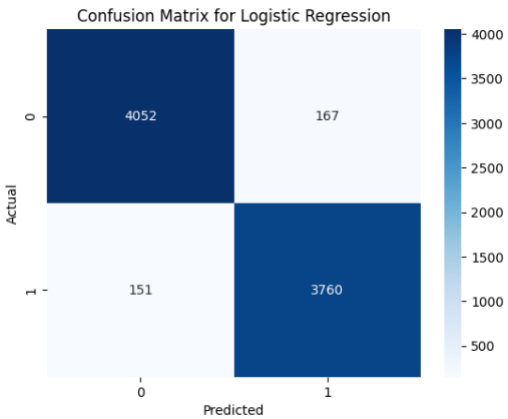
Few misclassification rates – only 3.63% of cases incorrectly classified, SVM is one of the most accurate models so far.	

3.5 *Logistic Regression*

Table 11a.
Performance metrics of Logistic Regression

	Precision	Recall	F1-score	Support	Overall Accuracy	Overall AUC
No stroke	0.96	0.96	0.96	4219	0.961	0.961
Stroke	0.96	0.96	0.96	3911		
Interpretation	Very strong performance with an accuracy of 96.09% and AUC of 0.961, great ability to distinguish between both classes. Precision, recall and F1-scores are identical between both classes, indicating its ability to minimise false predictions.					

Figure 15.
Confusion matrix visualisation and table



Of 8130 cases, the logistic regressor accurately predicted 7812 (96.09%) cases while 318 (3.91%) were misclassified, including 167 false positives and 151 false negatives. These values are relatively low, proving this model exceeded in minimising false classifications.

Number of instances	True NS	False NS	False S	True S	Total Correct Predictions	% of correct predictions	Total incorrect Predictions	% of incorrect predictions
8130	4052	167	151	3760	7812	96.09%	318	3.91%

Table 11b.
Evaluation of Logistic Regression

Strengths	Limitations
High Accuracy and AUC – accuracy of 96.09% and AUC of 0.961 show balanced and excellent performance.	**False negatives –** even though it is low, 167 false negatives can be critical in healthcare settings, where actual strokes are being predicted as no strokes.

Consistent metrics – performance metrics of precision, recall and F1-score were all 0.96, highlighting a perfectly balanced performance in distinguishing between both classes.	**Assumption of linearity –** this model assumes there is a linear relationship between features, making it less flexible for where there is complexity.
Simplicity – easy to implement the logistic regressor and to interpret it, ideal for healthcare settings	

3.6 *AdaBoost Classifier*

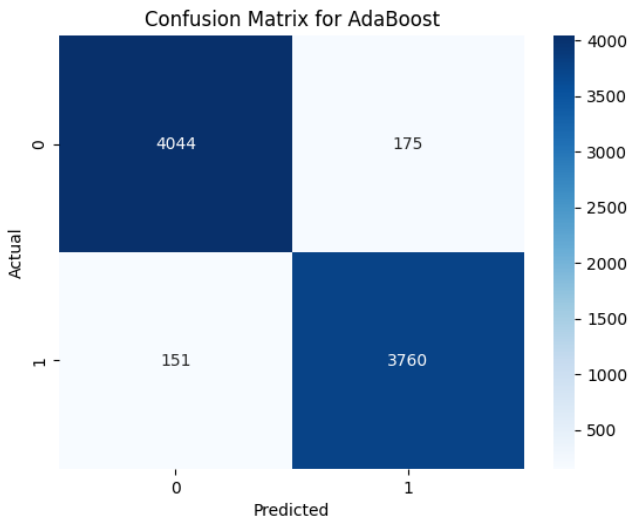
Table 12a.

Performance metrics of AdaBoost classifier

	Precision	Recall	F1-score	Support	Overall Accuracy	Overall AUC
No stroke	0.96	0.96	0.96	4219	0.960	0.960
Stroke	0.96	0.96	0.96	3911		
Interpretation	This final classifier also demonstrates an exceptional performance with high accuracy (0.960) and discriminatory power between both classes (AUC = 0.960). Precision, recall and F1-scores are consistent and well balanced throughout both classes with a value of 0.96, minimising false predictions of either class.					

Figure 16.

Confusion Matrix visualisation and table



Out of 8130 cases, 7804 instances were correctly classified (95.99%) while 326 cases (4.01%) remained misclassified: 175 false positives (S predicted as NS) and 151 false negatives (NS predicted as S).

Number of instances	True NS	False NS	False S	True S	Total Correct Predictions	% of correct predictions	Total incorrect Predictions	% of incorrect predictions
8130	4044	151	175	3760	7804	95.99%	326	4.01%

Table 12b.

Evaluation of AdaBoost classifier

Strengths	Limitations
High Accuracy and AUC – accuracy of 95.99% and AUC of 0.960 shows balanced and excellent performance.	**False negatives –** even though it is low, 151 false negatives can be critical in healthcare settings, where actual strokes are being predicted as no strokes.
Consistent metrics – performance metrics of precision, recall and F1-score were all 0.96, highlighting a perfectly balanced performance in distinguishing between both classes.	**Sensitive to noisy data –** this model is sensitive to noisy data and outliers, increasing the likelihood of overfitting if the data is not pre-processed.
Ensemble learning – the findings of AdaBoost demonstrates the effectiveness of ensemble learning, which combines multiple models to improve the overall performance of classification in this dataset, in this instance, DT was used as a baseline model.	

3.7 Comparison

Table 13.
Comparison table for all models

	Decision Tree	Naïve Bayes	K-Nearest Neighbour	Support Vector Machine	Logistic Regression	AdaBoost
Precision NS	0.96	0.96	0.96	0.96	0.96	0.96
Precision S	0.92	0.96	0.85	0.96	0.96	0.96
Recall NS	0.93	0.97	0.84	0.97	0.96	0.96
Recall S	0.95	0.96	0.97	0.96	0.96	0.96
F1-Score NS	0.94	0.97	0.90	0.97	0.96	0.96
F1-Score S	0.94	0.96	0.90	0.96	0.96	0.96
Overall Accuracy	0.941	0.964	0.899	0.963	0.961	0.960
AUC	0.940	0.964	0.902	0.960	0.960	0.960
Number of instances	8310	8310	8310	8310	8310	8310
True NS	3910	4078	3530	4075	4052	4044
False NS	176	151	129	151	151	151
False S	309	141	689	144	167	175
True S	3735	3760	3782	3760	3760	3760
Total Correct Predictions	7645	7838	7312	7835	7812	7804
Percentage of correct predictions	94.03%	96.40%	89.94%	96.37%	96.09%	95.99%

Total incorrect predictions	485	292	818	295	318	326
Percentage of incorrect predictions	5.97%	3.60%	10.06%	3.63%	3.91%	4.01%

Between all six models of classification, there are very relatively small differences in performance. Naïve Bayes demonstrated the best classification performance, with the highest accuracy of 96.40%, while ever so slightly outperforming other models in precision, recall and F1-scores, but the margins of these differences are very minimal. SVM, Logistic Regression and AdaBoost models closely followed Naïve Bayes in classification performance, all revealing overall accuracies around 96%. Decision Tree slightly underperformed with a lower accuracy of 94.1% but remains an impressive score, this method is still suitable as a reliable classification method. Although, the KNN classifier performed the worst with the lowest accuracy of 89.90%, this is still a high score, but relative to the other model's performance, KNN demonstrated the weakest performance overall.

In this instance, any of the top performing models would be the most suitable to use, including, Naïve Bayes, SVM, Logistic Regression and AdaBoost. Given they all exhibited almost identical classification abilities, the choice on the most suitable technique depends on other factors such as computer efficiency, which excludes SVM given that it is computationally expensive.

4. Regression

Regression was performed on the BMI variable, referring to the respondent's body mass index. The original numerical dataset was not used in this task since the resampled dataset that address class imbalance produced better more accurate results. Linear Regression (LiR), Regression Tree (RT) and K-Nearest Neighbour Regression (KNNR) were applied. The strength of the model's predictions was evaluated by metrics including, mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), R^2 score and adjusted R^2.

4.1 *Linear Regression*

Table 14a.
Performance metrics for LiR

Model	MAE	MSE	RMSE	R^2	Adjusted R^2
Linear Regression	3.85	25.32	5.03	0.147	0.143

An MAE value of 3.85 shows, on average, the model's BMI value predictions deviate from the actual values by 3.85 units. A relatively large MSE (25.32) emphasises these deviations. The RMSE (5.03) further reflects these deviations, representing larger errors in some predictions. R^2 (0.147) means the LiR model can only explain 14.7% of variance in BMI values; the Adjusted R^2

value of 14.3% is only slightly lower than the R^2, therefore, adding predictors failed to improve the LiR's performance.

Table 14b.
Evaluation of LiR

Strengths	Limitations
Simplicity – operates on linearity between predictor and target variables, making it easy to explain how predictors influence the target variable.	**Assumption dependence –** a low R^2 value suggests linearity was not present in this context.
Computationally efficient – this model requires less processing power compared to other models, even with large datasets	**Not Flexible –** Linear regression is limited to linear relationships, therefore, struggles with non-linear relationships.
Helpful assumptions – assumptions of linearity, independence or normality allow easy detection of violations to these assumptions, quickly highlighting potential issues in the data or model.	

4.2 <u>*Decision Tree Regression*</u>

Table 15a.
Performance metrics of DTR

Model	MAE	MSE	RMSE	R^2	Adjusted R^2
Decision Tree Regression	3.51	22.07	4.70	0.256	0.253

An MAE of 3.51 suggests the DTR model's predictions deviate 3.51 units from the actual values, an improvement compared to the LiR's MAE of 3.85. RMSE of 4.70 is slightly better than that of LiR's model, implying the DTR is slightly better at minimising errors, nevertheless there is room for improvement. MSE of 22.07 suggests a lot of overall error, but it is lower compared to LiR. The DTR accounts for 25.6% of variance in BMI values, slightly outperforming LiR, but like LiR, adjusting the number of predictors did not improve performance, given a similar R^2 and adjusted $R^{2.}$

Table 15b.
Evaluation of DTR

Strengths	Limitations
Easy to interpret – Decision Trees mimic human decision making processes, making it easier to understand.	**Overfitting –** tend to overfit the data
Handles non-linear relationships – unlike LiR, therefore able to perform for more complex data.	**Bias –** this model favours features with more levels, leading to bias towards the majority class, so data must be balanced before conducting DTR.

Large datasets – can efficiently and effectively handle large datasets, therefore, more suitable for real-world scenarios.	

4.3 *K-Nearest Neighbour*

Table 16a.
Performance metrics for KNN

Model	MAE	MSE	RMSE	R^2	Adjusted R^2
K-Nearest Neighbour	2.87	20.49	4.53	0.310	0.307

On average, the KNN model's predictions deviates 2.87 units from the actual values, outperforming both LiR (MAE = 3.85) and DTR (MAE = 3.51). The RMSE for KNN (4.53) is also lower than that of DTR and LiR, suggesting KNN performs better at minimising large errors. The MSE of 20.49 is still relatively high suggesting there is room for improvement, nevertheless, outperforms DTR and LiR in minimising overall error. KNN can explain 31% of variance in BMI values, a distinct improvement over LiR (14.7%) and DTR (25.6%), indicating KNN better identifies underlying patterns in the data. Although, there is little difference between R^2 and adjusted R^2, so additional predictors did not substantially improve KNN's performance.

Table 16b.
Evaluation of KNN

Strengths	**Limitations**
Simplicity – easy to implement KNN, its predictions do not require complex mathematical models	**Lack of interpretability -** unlike LiR and DTR, KNN cannot provide information about the relationships between features and target variables.
Non-Parametric – KNN does not assume a specific relationship between features and target variables, deeming it suitable for non-linear data and data with an unknown distribution	**Choice of *k* -** difficult to select the right value for *k,* too small can cause overfitting while too large can introduce bias.
Adaptability – adjusting the number of neighbours (k) allows the model to balance bias and variance, making it flexible for different datasets.	

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4.4 *Support Vector Regression*

Table 17a.
Performance metrics for SVR

Model	MAE	MSE	RMSE	R^2	Adjusted R^2
Support Vector Regression	3.49	22.61	4.75	0.238	0.235

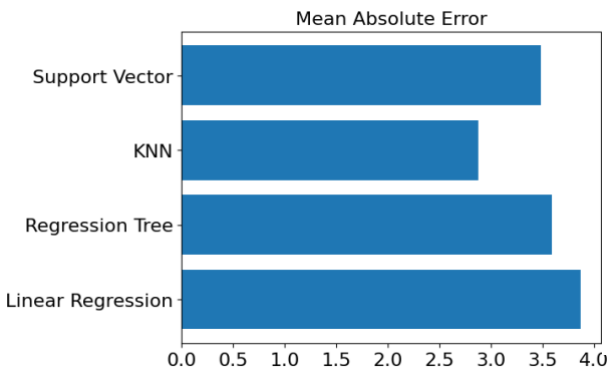
SVR's predictions deviate by 3.49 units from the actual values, on average, outperforming LiR(3.85) and DTR (3.51), but it is not as precise as KNN (2.87). The RMSE of 4.75 suggests larger errors in some of SVR's predictions compared to KNN (4.53), nevertheless, it is an improvement on both LiR (5.03) and DTR (4.70). An MSE of 22.61 highlights minimised overall error compared to LiR (25.32) and DTR (22.07), but not better than KNN (20.49). The R2 value shows SVR can explain around 23.8% of variance in BMI, better than LiR (14.7%) although it underperforms compared to KNN (31%) and DTR (25.6%). Adjusted R^2 value is slightly lower than the R^2 value, indicating very little effect on adding predictors to the model.

Table 17b

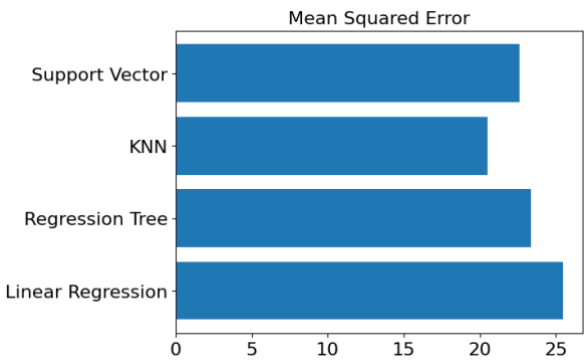
Evaluation of SVR

Strengths	**Limitations**
Effective in High-Dimensional spaces – works well with datasets that have many features, making it ideal for complex problems.	**Computationally expensive –** SVR requires a lot of computational power, especially with large datasets.
Reduces risk of overfitting – maximises the margin between data points and decision boundaries	**Less interpretable –** SVR doesn't provide information about the relationship between predictors and target variables, so it is less interpretable than other models such as LiR and DTR.
Flexibility – use of kernels allows SVR to handle non-linear relationships, therefore, can model complex patterns	

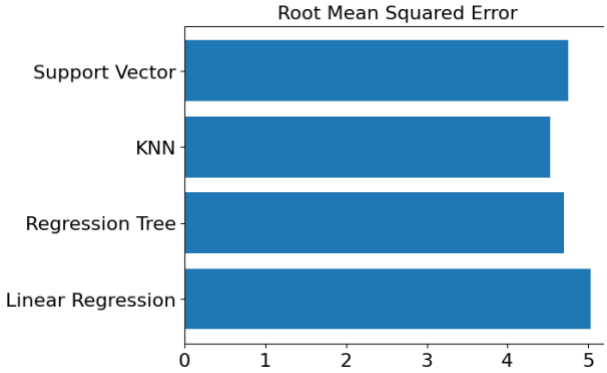
4.5 **Comparison**



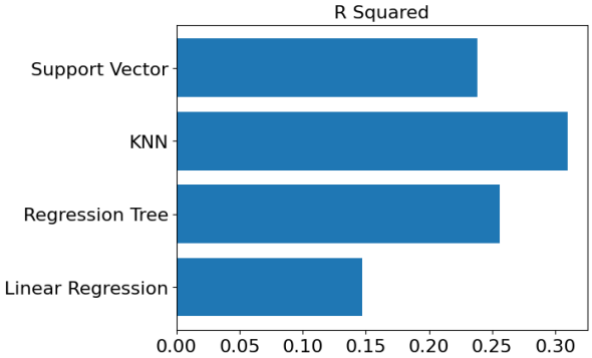
KNN predicted true BMI values the most accurately, LiR performed the worst, while SVR and DTR performed moderately.



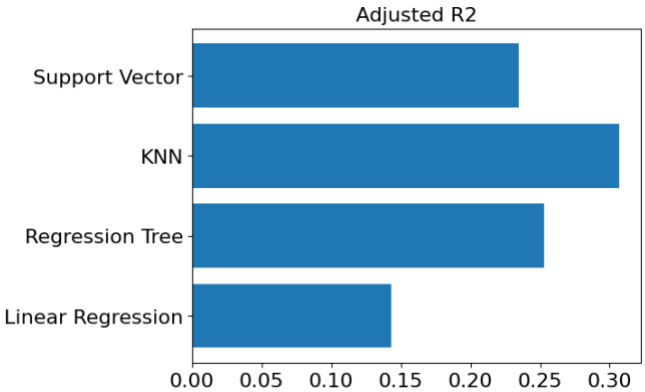
KNN best accounted for overall errors, while LiR showed greatest difficulty in minimising overall error.



KNN handles larger errors the best, only slightly outperforming the regression tree and support vector regressor, linear regression performs moderately.



KNN accounts for the highest proportion of variance in BMI values, while LiR explains the least variance. SVR performs better than DTR.



KNN remains the strongest in adjusted R2, confirming its robustness in generalising true patterns in the data to new, similar data points with added predictors. SVR follows KNN, demonstrating good generalisation but outshined by KNN. DTR shows some capacity to generalise but is outperformed by KNN and SVR. LiR performs the worst, confirming it received very little benefit from adding predictors.

Overall, KNN outperforms all the models across all metrics, making it the most effective regression model for this dataset. SVR and DTR performed reasonably well, though, SVR had the slight edge with lower MSE and RMSE values, making SVR the second-best effective model and a reasonable alternative for more complex and high-dimensional relationships, and DTR the penultimate one. LR underperformed across all metrics, deeming it the least suitable for this dataset.

5. Clustering

This section will explore the application of three clustering algorithms: K-Means clustering, Hierarchical Clustering and Gaussian Mixture Model (GMM) on the dataset. The analysis will focus on understanding the impact of dataset demographics and health indicators on health conditions.

5.1 *K-Means Clustering*

Table 18a.
K-Means clusters

Variable	Cluster 1 (standardised)	Cluster 1 (variable)	Cluster 2 (standardised)	Cluster 2 (variable)
Gender	-0.02886381	Female	0.0670361	Female

Age	0.49744068	54.48	-1.15530436	17.10
Hypertension	0.13865461	No	-0.32202488	No
Heart Disease	0.10040656	No	-0.2331939	No
Ever Married	0.58950204	Yes	-1.36911658	Yes
Work Type	-0.27453031	Private	0.63759574	Self-employed
Residence Type	0.00915018	Rural	-0.02125126	Rural
Glucose	0.11587275	111.39	-0.2691141	93.96
BMI	0.2753999	31.01	-0.63961537	23.96
Smoking status	0.20999589	Never Smoked	-0.48771478	Formerly Smoked
Stroke	0.09354905	No	-0.21726736	No
Interpretation	This algorithm identified distinct clusters based on demographics such as age, and health indicators like glucose and BMI values, revealing distinct health-related patterns, for instance, younger individuals form clusters with lower glucose and BMI levels, while older individuals form clusters with higher glucose and BMI values – key risk factors for health conditions.			

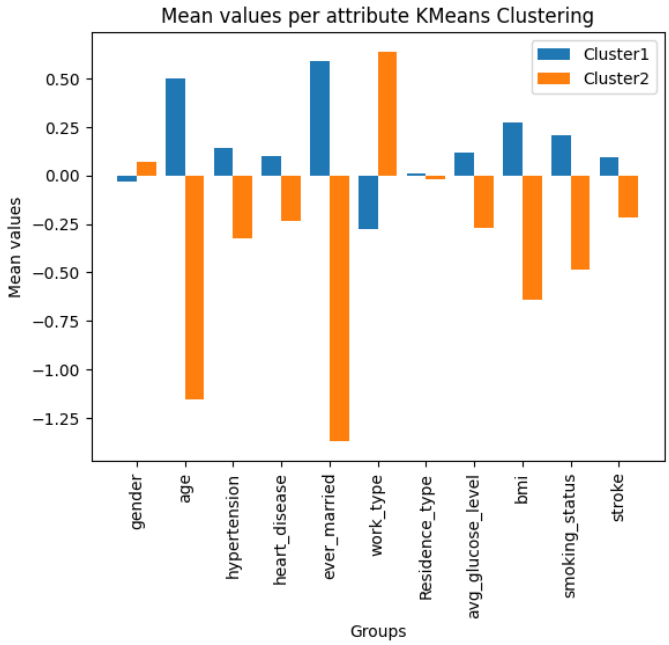


Table 18b:

Evaluation of K-Means Clustering

Strengths	**Weaknesses**
Fast and Efficient: works quickly, great for large datasets	**Performance limitations:** where groups share traits or overlap, K-Means struggles to handle this well.
Interpretability: K-Means groups items based on their similarity, a relatively straightforward technique, making this method easy to utilise and explain	**Ignores shape/size of groups:** This model assumes the shapes of all groups are round/spherical and the same size, which isn't likely to always reflect reality. For instance, a group may be large and spread

	out while another is smaller and less widespread.
Flexible for many uses: great at grouping people based on similar characteristics, in this instance, such as demographic information (age) and health indicators (glucose and BMI).	

5.2 Hierarchical Clustering

Table 19a.
Hierarchical clusters

Variable	Cluster 1 (standardised)	Cluster 1 (variable)	Cluster 2 (standardised)	Cluster 2 (variable)
Gender	1.18807255	Male	1.18807255	Male
Age	-0.67343309	28.00	1.27257113	72.00
Hypertension	-0.32860186	No	-0.32860186	No
Heart Disease	-0.2389468	No	4.18503199	Yes
Ever married	-1.3814364	Yes	0.72388421	Yes
Work Type	-1.98838433	Government Job	0.76343762	Self-employed
Residence Type	0.98407974	Urban	-1.01617781	Urban
Glucose	0.27633753	118.66	-0.70203817	74.36
BMI	0.4435762	32.30	-0.20599748	27.30
Smoking Status	0.58155233	Never Smoked	0.58155233	Never Smoked
Stroke	-0.22632726	No	-0.22632726	No
Interpretation	Hierarchical clustering captured more nuanced associations such as between heart disease and work type, which were less distinct in other models. Self-employed work type formed clusters with history of heart disease while health predictors like BMI, glucose and age were not indicative of heart disease, and any other factors remained constant between both clusters. This suggests self-employment has a potential association with heart disease.			

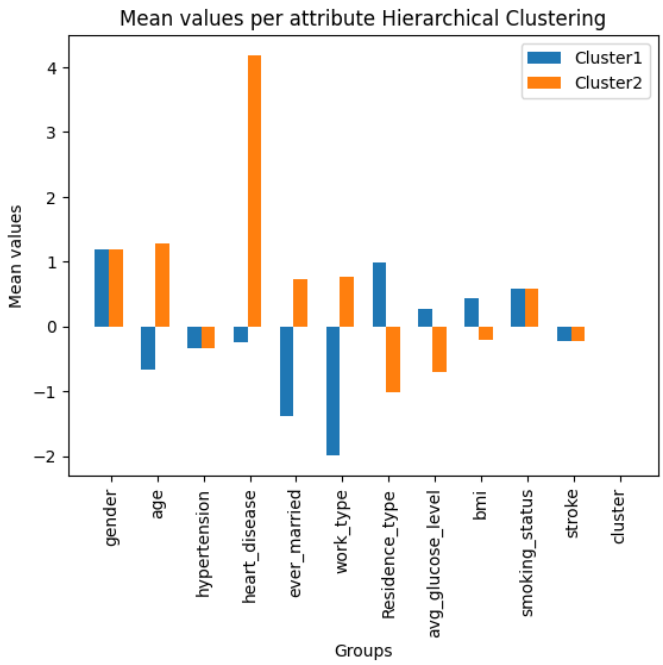


Table 19b.

Evaluation of hierarchical clustering

Strengths	**Weaknesses**
Works for any group shape: can handle clusters of different shapes and sizes, not just round shapes, unlike K-Means.	**Computationally slow:** this algorithm compares all values against every other value, for big datasets, this will take a lot of time and computer power.
Works well for small to medium data: very strong performance in smaller datasets.	

5.3 *Gaussian Mixture Model*

Table 20a.

GMM Clusters

Variable	Cluster 1 (standardised)	Cluster 1 (variable)	Cluster 2 (standardised)	Cluster 2 (variable)
Gender	0.06647752	Female	-0.15234306	Female
Age	-1.11603978	17.99	0.42888881	52.92
Hypertension	-0.28893365	No	0.00660615	No
Heart Disease	0.2389468	No	0.04563663	No
Ever Married	-1.3814364	Yes	0.57614241	Yes
Work Type	0.53681699	Self-employed	-0.18065724	Private
Residence Type	-0.02781525	Rural	-0.0803848	Rural
Glucose	-0.08764525	102.18	0.1041851	110.87
BMI	0.76478369	34.77	0.2584287	30.88

Smoking Status	0.50550615	Never Smoked	0.18857	Never Smoked
Stroke	0.22632726	No	0.04529309	No
Intepretation	GMM offered more nuanced insights into relationships within the data. It further highlighted the importance of demographic factors such as work type and age in building health-related clusters. Cluster 1 and 2 shared similar variables across clusters, but this model performed well at handling overlapping features, such as in glucose levels and BMI.			

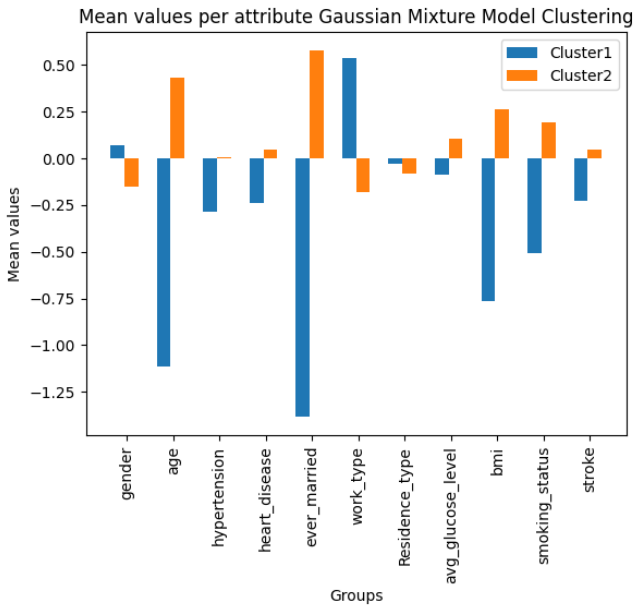


Table 20b.

Evaluation of GMM

Strengths	**Weaknesses**
Overlapping groups: GMM performs well in handling clusters that overlap in different variables.	**Requires tuning:** initialisation and parameters need to be set beforehand, difficult to do if you do not know the data well.
More versatile: GMM can identify and understand even more complex shapes and sizes of groups.	**Computationally intensive –** requires a lot of computational power with large datasets

Analysis: *Impact of Demographics and Health Indicators on Health Conditions*

Demographics and Clusters

- **Age:** Clusters of older age tended to cluster with higher BMI, glucose levels and health conditions, indicating age is a significant risk factor compromising health.
- **Gender:** gender differences across clusters in all the models showed clustering patterns linked with health outcomes, for instance, cluster 2 in hierarchical clustering formed

clusters between males and individuals with history heart disease as opposed to clusters in K-Means and GMM which formed clusters between females and no heart disease.

- **Work type and residence type:** Work type was differentiated between clusters effectively, where self-employed individuals form a distinct group. Residence type was less effective but clusters were distinguished across hierarchical clustering, highlighting urban groups, while K-Means and GMM highlight the rural population.

Health Indicators and Clusters

Only cluster 2 in hierarchical clustering captured an individual with heart disease.

- **BMI:** BMI was moderate in this cluster so no association between BMI and heart disease development can be made.
- **Glucose Levels:** Similarly to BMI, glucose level remained moderate/low so it cannot be assumed glucose level had any effect in developing heart disease.
- **Smoking status:** history of smoking formed clusters with young individuals, indicating potential early health risk.

Health conditions across clusters

No clusters were formed for those with a history of hypertension or having a stroke, but the hierarchical model reported a cluster of an individual with heart disease. Although, no health predictors such as age, BMI, glucose levels directly associated with hypertension, heart disease or stroke.

The only conclusion that can be made from this analysis is the increase in BMI and average glucose levels when ageing. This analysis failed to prove the influence of demographic factors and health indicators on health conditions.

Conclusion

Hierarchical clustering provided the most comprehensive findings, particularly in exploratory settings, offering insights with distinct patterns. GMM addressed nuances in the data with overlapping clusters, while K-Means performed well in initial assessments of clusters. Each clustering offered a unique insight, but hierarchical clustering emerged as the most effective. The overall results helped foster practical implications including the use of methods to target groups at risk based on age, BMI and glucose levels.

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